Week 2 question sheet

Problem sheet

Instructions

Attempt all ten questions and all 17 scorable parts. The target time is 240 minutes and the total is 100 marks. No calculator or software is needed. Show row operations, intermediate probabilities and any condition used to justify a classification. Give exact values unless a question says otherwise.

Use hints.md only after a serious attempt. Do not consult solutions.md until you have completed or paused every part.

Question 1: matrix multiplication (M4, 11 marks, 26 minutes)

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 3 & 2 \end{bmatrix}.$$

Let $\mathbf{u} = [1, -1]^{\mathsf{T}}$.

(a) `W02-Q01-PA` — **5 marks.** Compute $\mathbf{AB}$, stating the dimension of the product. Then compute both $\mathbf{A}(\mathbf{Bu})$ and $(\mathbf{AB})\mathbf{u}$ to verify associativity for these matrices and this vector.

(b) `W02-Q01-PB` — **6 marks.** Compute $\mathbf{BA}$. Verify directly that $(\mathbf{BA})^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}$. Use your results and a dimension argument to explain why matrix multiplication is not commutative in this example.

Question 2: determinant, inverse and solution (M4, 12 marks, 26 minutes)

Let

$$\mathbf{C} = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}.$$

(a) `W02-Q02-PA` — **6 marks.** Compute $\det(\mathbf{C})$ and $\mathbf{C}^{-1}$. Verify the inverse by computing $\mathbf{C}\mathbf{C}^{-1}$.

(b) `W02-Q02-PB` — **6 marks.** Use the inverse to solve the two systems simultaneously in the matrix equation

$$\mathbf{CX} = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix},$$

and verify the result by multiplying $\mathbf{CX}$. State why each right-hand side has a unique solution.

Question 3: Gaussian elimination (M4, 10 marks, 22 minutes)

`W02-Q03-PA` — **10 marks.** Use Gaussian elimination to solve

$$\begin{aligned} x + y + z + w &= 5, \\ 2x - y + z &= -1, \\ x + 2y - z + 2w &= 12, \\ 3x + y + 2z - w &= 0. \end{aligned}$$

Show the row operations and classify the system as having a unique solution, no solution or multiple solutions. Verify your solution in the original four equations.

Question 4: rank and parameter-dependent solvability (M4, 15 marks, 30 minutes)

For $k \in \mathbb{R}$, consider

$$\begin{aligned} x + y + z &= 1, \\ 2x + 3y + 4z &= 2, \\ 3x + 4y + 5z &= k. \end{aligned}$$

Let $\mathbf{A}$ be the coefficient matrix and let $\mathbf{b} = [1, 2, k]^{\mathsf{T}}$.

(a) `W02-Q04-PA` **— 7 marks.** Row-reduce the augmented matrix far enough to determine $\operatorname{rank}(\mathbf{A})$ and $\operatorname{rank}([\mathbf{A} \mid \mathbf{b}])$ when $k = 3$ and when $k \neq 3$.

(b) `W02-Q04-PB` **— 8 marks.** For each case, classify the system and justify the classification using the ranks and the number of unknowns. When $k = 3$, write the solution set using one free parameter and verify that the parameterisation satisfies all three original equations.

Question 5: a rank claim (M4, 10 marks, 18 minutes)

`W02-Q05-PA` **— 10 marks.** A student claims that for every pair of 2×2 matrices $\mathbf{P}$ and $\mathbf{Q}$,

$$\operatorname{rank}(\mathbf{PQ}) = \min\{\operatorname{rank}(\mathbf{P}), \operatorname{rank}(\mathbf{Q})\}.$$

Construct two rank-one 2×2 matrices for which $\mathbf{PQ}$ is the zero matrix but $\mathbf{QP}$ is non-zero. Compute both products and the ranks of $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{PQ}$ and $\mathbf{QP}$. Explain both why your example disproves the claim and why the order of the factors matters here.

Question 6: conditioning within a cohort (S2, 11 marks, 27 minutes)

Let E be the event that a student has an economics background and let M be the event that the student completed a matrix refresher. Suppose

$$\mathbb{P}(E) = \frac{3}{5}, \qquad \mathbb{P}(M \mid E) = \frac{2}{3}, \qquad \mathbb{P}(M \mid E^c) = \frac{1}{4}.$$

(a) `W02-Q06-PA` **— 5 marks.** Construct the complete 2×2 probability table for the four events $E \cap M$, $E \cap M^c$, $E^c \cap M$ and $E^c \cap M^c$. Include the row and column marginal probabilities and check that the four joint probabilities sum to one.

(b) `W02-Q06-PB` **— 6 marks.** Compute both $\mathbb{P}(E \mid M)$ and $\mathbb{P}(E \mid M^c)$. Compare them with $\mathbb{P}(E)$, determine whether E and M are independent, and interpret the direction of the association in one sentence.

Question 7: total probability and Bayes' rule (S2, 11 marks, 29 minutes)

Students are assigned to review sessions R_1, R_2 and R_3 with probabilities $1/2$, $1/3$ and $1/6$, respectively. These events form a partition of the student group. Let C be the event that a student submits a correct worksheet. Suppose

$$\mathbb{P}(C \mid R_1) = \frac{3}{4}, \qquad \mathbb{P}(C \mid R_2) = \frac{1}{2}, \qquad \mathbb{P}(C \mid R_3) = \frac{1}{4}.$$

(a) `W02-Q07-PA` **— 5 marks.** Construct the joint probability table with rows R_1, R_2, R_3 and columns C, C^c, including marginal totals. Use the table or the law of total probability to compute $\mathbb{P}(C)$.

(b) `W02-Q07-PB` — **6 marks.** Given that a student's worksheet is correct, compute $\mathbb{P}(R_i \mid C)$ for all three sessions. Name the rule used, check that the three posterior probabilities sum to one, and identify the most likely session for a student with a correct worksheet.

Question 8: testing independence (S2, 6 marks, 14 minutes)

`W02-Q08-PA` — **6 marks.** Events A and B satisfy

$$\mathbb{P}(A) = \frac{1}{2}, \qquad \mathbb{P}(B) = \frac{2}{5}, \qquad \mathbb{P}(A \cup B) = \frac{7}{10}.$$

Compute $\mathbb{P}(A \cap B)$ and $\mathbb{P}(A \mid B)$. Determine whether A and B are independent using two equivalent probability equalities, and state whether the events are disjoint.

Question 9: interpreting a signal (S2, 6 marks, 18 minutes)

`W02-Q09-PA` — **6 marks.** Let D be an event with $\mathbb{P}(D) = 1/20$, and let S be the event that a signal is positive. Suppose $\mathbb{P}(S \mid D) = 9/10$ and $\mathbb{P}(S \mid D^c) = 1/10$. Compute both $\mathbb{P}(D \mid S)$ and $\mathbb{P}(D \mid S^c)$ using Bayes' rule. Compare the two posterior probabilities and explain why $\mathbb{P}(D \mid S)$ is not $9/10$.

Question 10: Week 1 retrieval (8 marks, 30 minutes)

These three parts retrieve previously studied material. Complete them without looking back at Week 1 notes.

(a) `W02-Q10-PA` — **M2, 3 marks.** Solve

$$\log x + \log(x - 3) = \log 4,$$

stating the domain restriction and rejecting any inadmissible root.

(b) `W02-Q10-PB` — **M3, 3 marks.** Find the second-order Taylor approximation of $f(x) = \log x$ about $x = 1$, and use it to approximate both $\log(0.9)$ and $\log(1.1)$. Explain why the quadratic correction has the same sign in both approximations.

(c) `W02-Q10-PC` — **M1 and S1, 2 marks.** Two representatives are selected uniformly from five students, with order irrelevant. Two of the students are Alice and Benoit. Let A be the event that Alice is selected and B the event that Benoit is selected. Express "at least one is selected" and its complement in set notation, using De Morgan's law. Compute the probability by a complementary counting argument, then confirm it by a direct count that avoids double-counting.

Self-test questions

Set aside 25 minutes and attempt all four parts without the reading guide, worked examples, hints or exercise solutions. No calculator is needed. Show the condition or equality that justifies each classification. The self-test has 20 marks and does not contribute to a checkpoint or mock-examination score.

Questions

`W02-ST01-PA` — **rank and a free variable (`M4`; 5 marks; 7 minutes)** An augmented matrix has already been reduced to

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

State the coefficient and augmented ranks, classify the system, and write the complete solution as a column vector with a real free parameter.

`W02-ST02-PA` — **determinant, inverse and solution (`M4`; 5 marks; 6 minutes)** Let

$$\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 5 \\ 13 \end{bmatrix}.$$

Compute $\det(\mathbf{D})$, state why the system $\mathbf{Dx} = \mathbf{b}$ has a unique solution, and find and verify $\mathbf{x}$.

`W02-ST03-PA` — **conditioning and independence (`S2`; 5 marks; 6 minutes)** Suppose

$$\mathbb{P}(A) = \frac{1}{2}, \qquad \mathbb{P}(B) = \frac{2}{5}, \qquad \mathbb{P}(A \cap B) = \frac{1}{5}.$$

Compute $\mathbb{P}(A \mid B)$. Decide whether A and B are independent and whether they are disjoint, giving a valid equality or inequality for each conclusion.

`W02-ST04-PA` — **total probability and Bayes' rule (`S2`; 5 marks; 6 minutes)** Events G and G^c form a partition, with

$$\mathbb{P}(G) = \frac{1}{4}, \qquad \mathbb{P}(S \mid G) = \frac{3}{4}, \qquad \mathbb{P}(S \mid G^c) = \frac{1}{4}.$$

Compute $\mathbb{P}(S)$ and $\mathbb{P}(G \mid S)$. State why $\mathbb{P}(S \mid G)$ is not itself the requested posterior.